

Bayesian Deep Learning for Uncertainty Quantification

Abstract

Deep neural networks have orchestrated a paradigm shift across artificial intelligence, enabling unprecedented predictive accuracy in computer vision, natural language processing, and complex scientific simulations. However, standard deterministic neural networks are notoriously prone to generating overconfident and miscalibrated predictions, particularly when deployed in out-of-distribution environments or when learning from limited data. This structural deficiency presents a critical barrier to the deployment of artificial intelligence in high-stakes, safety-critical systems where recognizing model ignorance is equally as important as predictive accuracy. Bayesian Deep Learning offers a mathematically rigorous framework to overcome these limitations by framing neural network parameters and predictive mappings as probabilistic distributions rather than point estimates. This survey provides an exhaustive review of the contemporary Bayesian Deep Learning landscape, with a specific focus on Uncertainty Quantification. It systematically categorizes the theoretical foundations of aleatoric and epistemic uncertainty, followed by a comprehensive exploration of approximate inference techniques. These include Variational Inference, Stochastic Gradient Markov Chain Monte Carlo, Laplace Approximations, Stein Variational Gradient Descent, and generalized ensembling methods. Furthermore, this survey examines the latest advances in parameter-efficient Bayesian fine-tuning—such as Bayesian Low-Rank Adaptation—which have successfully scaled uncertainty quantification to massive foundation models and Large Language Models. Finally, the review addresses unresolved anomalies such as the Cold Posterior Effect and outlines the future trajectory of scalable, sampling-based inference architectures, providing researchers with a unified perspective on the pursuit of calibrated and trustworthy deep learning.

Introduction

The remarkable success of deep neural networks in supervised, unsupervised, and reinforcement learning paradigms is extensively documented across the computational sciences. Modern architectures possess the functional capacity to map highly complex, non-linear relationships in high-dimensional data, consistently achieving human-level or superhuman performance on benchmark datasets. Despite this empirical success, a troubling phenomenon has been consistently observed: modern, highly parameterized neural networks are often profoundly miscalibrated [1]. While neural networks from prior decades reliably produced confidence scores that matched their empirical accuracy, the integration of extreme depth, width, and modern normalization techniques has led contemporary networks to yield heavily overconfident predictions, even when fundamentally incorrect [1, 2].

This systemic overconfidence severely limits the safe deployment of deep learning models. In domains such as autonomous driving, automated medical diagnosis, financial forecasting, and robotic control, a model must possess the capacity to signal when it encounters edge cases or novel inputs. In such instances, control must be safely handed over to human operators or

downstream verification systems [3, 4]. Uncertainty Quantification addresses this operational requirement by computing mathematically rigorous confidence intervals over network outputs, enabling systems to express appropriate hesitation when operating beyond their training manifold [5, 6].

Bayesian Deep Learning fundamentally reframes the deep learning optimization problem to support Uncertainty Quantification natively. Instead of applying Maximum Likelihood Estimation or Maximum A Posteriori estimation to isolate a single, deterministic set of optimal weights, the Bayesian paradigm seeks to evaluate the entire posterior distribution over the parameter space given the training data [7, 8]. By marginalizing predictions over this posterior, Bayesian neural networks inherently integrate an infinite ensemble of plausible hypotheses. This marginalization gracefully down-weights the confidence of predictions on data regimes sparsely covered by the training set, allowing the model to naturally "know what it does not know" [4, 9].

Historically, the exact computation of the Bayesian posterior in deep networks has been analytically intractable due to the non-convex, high-dimensional nature of the loss landscape, which routinely comprises millions or billions of parameters. Consequently, the field of Bayesian Deep Learning has been defined by the pursuit of tractable approximate inference [10]. Over the past decade, methodological and hardware advancements have revitalized interest in highly scalable approximations, leading to a renaissance in Bayesian techniques applied not only to standard convolutional and recurrent networks but also to modern Large Language Models [11, 12, 13]. This survey rigorously classifies the methodologies designed to extract calibrated uncertainty from deep neural architectures, analyzes their computational tradeoffs, and investigates the mathematical nuances governing their efficacy across diverse deployment scenarios.

Foundations of Uncertainty Quantification

To systematically analyze Bayesian techniques, it is essential to establish a precise taxonomy of uncertainty within deep learning. Uncertainty in predictive modeling is generally decomposed into two primary components: aleatoric uncertainty and epistemic uncertainty [14, 15]. Properly disentangling these components is necessary for designing systems that can differentiate between inherent noise and a sheer lack of knowledge.

Aleatoric vs. Epistemic Uncertainty

Aleatoric uncertainty, frequently referred to as data uncertainty, stems from the inherent stochasticity, noise, or irreducible randomness present in the data-generation process [14, 16]. Examples include sensor noise in a captured image, measurement truncation errors, semantic ambiguity in natural language, and conflicting ground-truth labels assigned to identical input features. Crucially, aleatoric uncertainty cannot be reduced by simply gathering a larger volume of identical data [15, 17]. It is an inherent property of the environment. Within neural architectures, it is often modeled by predicting a variance parameter alongside the mean output. Aleatoric uncertainty is further subdivided into two categories. Homoscedastic uncertainty assumes constant observation noise across all input variables and is typically learned as a single, global parameter. Conversely, heteroscedastic uncertainty assumes that the variance of the observation noise is a dynamic function of the input data. Heteroscedastic

modeling is vital in tasks like computer vision depth regression, where featureless objects naturally produce more measurement noise than highly textured surfaces [14, 15]. Epistemic uncertainty, universally known as model uncertainty, arises from a lack of knowledge regarding the optimal model parameters. It is highly prominent in regions of the input space where training data is sparse or entirely absent, often referred to as out-of-distribution regions [14, 18, 19]. Because epistemic uncertainty reflects the ignorance of the model rather than inherent environmental noise, it is theoretically reducible by collecting more training data in the poorly sampled regions [15]. Bayesian Deep Learning primarily targets the rigorous quantification of epistemic uncertainty through the posterior parameter distribution, treating the weights themselves as random variables [8, 20].

Uncertainty Type	Source	Reducibility	Modeling Approach
Aleatoric (Homoscedastic)	Inherent noise, constant across data	Irreducible	Global learned variance parameter
Aleatoric (Heteroscedastic)	Inherent noise, varying across data	Irreducible	Input-dependent variance output layer
Epistemic	Lack of knowledge, out-of-distribution	Reducible via more data	Posterior distribution over model weights

The Bayesian Predictive Framework

In a standard classification or regression task, given a training dataset comprising input-output pairs, a deterministic neural network yields a point estimate for the parameters by minimizing a loss function. In contrast, the Bayesian framework computes the posterior over the weights via Bayes' theorem, updating a prior belief distribution based on the observed data likelihood [7, 21]. The evidence, or marginal likelihood, normalizes this distribution.

To make a prediction for a newly observed test point, a Bayesian model computes the posterior predictive distribution by marginalizing the likelihood over the entire parameter posterior. This integral guarantees that when the posterior is broad—indicating high epistemic uncertainty—the resulting predictive distribution will appropriately exhibit high entropy, thereby preventing overconfident classifications [21, 22]. The total predictive variance can subsequently be mathematically decomposed into the sum of the expected aleatoric variance and the variance of the expected mean, which neatly captures the epistemic component [14, 15].

Calibration and Proper Scoring Rules

An uncertainty estimate is only practically useful if it corresponds to the true statistical likelihood of correctness. A model is deemed perfectly calibrated if its confidence matches its expected empirical accuracy. For instance, if a calibrated model predicts an event with eighty percent confidence across a subset of data, those predictions should be empirically correct exactly eighty percent of the time [1, 23].

The primary metric used to quantify miscalibration is the Expected Calibration Error. This metric partitions the model's confidence scores into equally spaced bins and computes the weighted average of the absolute difference between the empirical accuracy and the mean confidence within each bin [24, 25]. Beyond Expected Calibration Error, proper scoring rules, such as the Negative Log-Likelihood and the Brier Score, are frequently utilized to evaluate predictive uncertainty during both training and evaluation, as their expected values are uniquely minimized by the true underlying data-generating distribution [23, 26, 27].

Methods

Given the high dimensionality of modern neural network parameters, exact computation of the posterior integral is mathematically intractable. Consequently, the literature on Bayesian Deep Learning comprises various approximation families, each navigating the tradeoffs between computational cost, scalability, and inference fidelity [10, 28]. This section details the predominant methodologies utilized to extract calibrated uncertainty.

Variational Inference

Variational Inference reformulates Bayesian marginalization as a deterministic optimization problem. Instead of directly computing the intractable posterior, Variational Inference posits a tractable family of parameterized distributions and optimizes these variational parameters to minimize the Kullback-Leibler divergence between the approximate distribution and the true posterior [28, 29].

Minimizing this divergence is mathematically equivalent to maximizing the Evidence Lower Bound. The Evidence Lower Bound comprises two distinct terms: a likelihood cost that ensures the chosen parameter distribution fits the training data well, and a complexity cost that acts as a regularizer, penalizing the variational distribution if it deviates significantly from the selected prior [29, 30].

The most ubiquitous implementation is Mean-Field Variational Inference, which makes the strong assumption that the variational distribution factorizes completely over the network weights, typically utilizing independent Gaussian distributions for each parameter [28]. The "Bayes by Backprop" algorithm popularized this approach by leveraging the reparameterization trick to compute unbiased Monte Carlo estimates of the Evidence Lower Bound gradients, thereby enabling end-to-end training via standard backpropagation [29, 30]. Under this framework, every network weight is represented by a mean and a variance, effectively doubling the total parameter count. This explicit modeling of variance enforces exploration during training and robustifies the network against noisy or adversarial data [30, 31].

The mean-field assumption, however, is highly restrictive, as it completely ignores the complex covariances and structural dependencies between weights across different layers. To address this, recent advancements have proposed relaxations using block-diagonal structures or

low-rank parameterizations [28]. Furthermore, researchers have successfully substituted standard Gaussian priors with spike-and-slab distributions. A spike-and-slab prior is a mixture of a highly concentrated Gaussian centered at zero (the spike) and a broad, diffuse Gaussian (the slab). Applying Variational Inference with spike-and-slab priors facilitates simultaneous structural sparsification and uncertainty quantification, natively penalizing over-parameterized paths in deep architectures and allowing the network to prune unnecessary connections dynamically during inference [31, 32].

Recognizing the immense computational overhead of optimizing a probability distribution over the entire parameter space of massive modern networks, Last-Layer Variational Inference restricts the probabilistic treatment entirely to the final linear layer of the network, retaining deterministic point estimates for the underlying deep feature extractor [33, 34]. This hybrid approach effectively matches the representational capacity of a full Bayesian linear model operating on a highly optimized learned feature space, delivering highly competitive epistemic uncertainty at a mere fraction of the computational and memory costs associated with full-network Variational Inference [33, 35].

Stochastic Gradient Markov Chain Monte Carlo

While Variational Inference scales well to large datasets, its representational capacity is strictly limited by the chosen variational family, meaning it cannot approximate highly multimodal or skewed posteriors accurately. Markov Chain Monte Carlo methods, conversely, aim for asymptotic exactness by generating a sequence of parameter samples that eventually converge to the true posterior distribution, regardless of its topological complexity [8, 36]. Traditional Markov Chain Monte Carlo requires calculating gradients over the entire dataset at every step, rendering it computationally infeasible for deep learning. Stochastic Gradient Markov Chain Monte Carlo overcomes this limitation by utilizing mini-batch gradient estimators [37, 38].

The foundational algorithm in this space is Stochastic Gradient Langevin Dynamics, which elegantly combines stochastic gradient descent with Langevin diffusion dynamics [37, 39]. The discrete-time parameter update rule injects carefully calibrated Gaussian noise into the optimization trajectory. Initially, when the step size is relatively large, the gradient term dominates, allowing the algorithm to rapidly search for a high-probability mode much like standard stochastic gradient descent. As the learning rate decays toward zero, the injected Gaussian noise begins to dominate the stochastic gradient variance, and the trajectory transitions seamlessly into an exact sampling phase, exploring the local posterior landscape to generate valid samples from the true distribution [37, 40].

To accelerate exploration through pathological curvature and high-dimensional parameter spaces, researchers introduced Stochastic Gradient Hamiltonian Monte Carlo. This algorithm incorporates auxiliary momentum variables, mirroring the physical dynamics of a particle moving through a potential field with friction, which serves to counteract the disruptive noise inherent in mini-batch gradient estimation [41, 42]. Further advances include Cyclical Stochastic Gradient Markov Chain Monte Carlo, which repeatedly raises and lowers the learning rate schedule. This cyclical injection of energy allows the Markov chain to escape local modes and capture a diverse set of samples across multiple isolated basins of attraction, significantly

improving the representation of multimodal posteriors [43, 44].

Modern software frameworks leverage advanced hardware compilation to parallelize multiple Markov Chain Monte Carlo chains effortlessly, significantly lowering the barrier to entry for sampling methods. The field is increasingly recognizing that with modern parallelization techniques and low-precision gradient accumulators, sampling methods are achieving computational parity with deterministic training, prompting a paradigm shift asserting that sampling-based inference represents the definitive future of Bayesian Deep Learning [45, 46].

Stein Variational Gradient Descent

Bridging the gap between the parametric constraints of Variational Inference and the computational demands of Markov Chain Monte Carlo, Stein Variational Gradient Descent offers a deterministic particle-based approach to Bayesian inference [47, 48]. Stein Variational Gradient Descent initializes a set of interconnected particles (representing diverse neural network weight configurations) and iteratively transports them to match the target posterior distribution.

The algorithm applies a form of functional gradient descent that minimizes the Kullback-Leibler divergence within a Reproducing Kernel Hilbert Space. By leveraging a repulsive kernel function, Stein Variational Gradient Descent forces the particles to spread out and cover the entire posterior landscape, preventing them from collapsing into a single Maximum A Posteriori mode [47, 49]. This Wasserstein gradient flow guarantees that the empirical distribution of the particles asymptotically matches the true posterior.

Recent advancements have enhanced this framework by substituting scalar-valued kernels with matrix-valued kernels, allowing the algorithm to incorporate geometric preconditioning matrices, such as the Fisher information matrix, thereby accelerating convergence in pathological probability landscapes [50, 51]. However, in exceedingly high-dimensional neural networks, Stein Variational Gradient Descent can suffer from variance collapse, wherein the particles fail to spread adequately across the probability space, leading to underestimated epistemic uncertainty [52]. Solutions such as message-passing Stein Variational Gradient Descent and neural-parameterized witness functions aim to circumvent this curse of dimensionality [52, 53].

Laplace Approximations

The Laplace Approximation offers a highly efficient, post-hoc technique for rendering a pre-trained deterministic neural network into a fully Bayesian model without necessitating complex modifications to the original training pipeline [54, 55]. The Laplace Approximation centers a multivariate Gaussian distribution directly around the Maximum A Posteriori estimate achieved via standard gradient descent optimization.

The covariance of this Gaussian is defined as the inverse Hessian matrix of the negative log-posterior evaluated at the Maximum A Posteriori estimate. The Hessian fundamentally describes the local curvature of the loss landscape [56, 57]. If the loss surface is remarkably sharp, denoting high curvature, the resulting variance is small; conversely, if the loss basin is flat, the variance is large, accurately reflecting higher epistemic uncertainty surrounding the optimal parameter configuration [57].

The exact Hessian matrix requires dimensions proportional to the square of the total parameter count, which is computationally intractable for networks comprising millions of weights. To render the Laplace Approximation practical, the Hessian must be structurally approximated. The Kronecker-factored Approximate Curvature leverages the inherent mathematical structure of neural network layers to decompose the massive Hessian into a Kronecker product of significantly smaller matrices, representing input covariances and output gradient covariances respectively [58, 59]. Modern deep learning libraries provide streamlined access to diagonal, block-diagonal, and Kronecker-factored curvature approximations, drastically minimizing memory overhead and enabling the rapid deployment of Laplace methods in production environments [54, 55].

While the Laplace Approximation successfully parameterizes a posterior distribution over network weights, propagating this uncertainty through deep, highly non-linear architectures remains mathematically challenging. Monte Carlo sampling of the Laplace posterior is standard practice but incurs a substantial computational penalty during inference [60, 61]. An elegant alternative is Local Linearization, also known as the Generalized Gauss-Newton predictive approximation. By executing a first-order Taylor expansion of the neural network around the Maximum A Posteriori estimate, the predictive distribution becomes analytically tractable [60, 62]. This transforms the network into a locally linear model, allowing a single forward pass to compute the predictive variance natively utilizing the Jacobian of the network output [62, 63].

Method	Approach	Deterministic/Stochastic	Inference Overhead
SGLD	Injects noise into gradients	Stochastic	High (requires sampling)
SVGD	Evolves interacting particles	Deterministic	High (multiple networks)
Laplace (Full)	Fits Gaussian via Hessian	Deterministic (Post-hoc)	Low to Medium
Laplace (Linearized)	Taylor expansion around MAP	Deterministic	Very Low (single pass)

Deep Ensembles and Monte Carlo Dropout

Despite the theoretical elegance of explicitly parameterizing Bayesian posteriors, pragmatic approximations such as Deep Ensembles and Monte Carlo Dropout have emerged as highly robust, easily implementable alternatives that consistently yield state-of-the-art empirical calibration and out-of-distribution detection. In a seminal contribution to the field, researchers demonstrated a profound theoretical

equivalence between standard neural networks equipped with dropout and an approximation to Deep Gaussian Processes [64, 65]. When dropout is applied sequentially before every weight layer, sampling from the network at inference time with dropout held active simulates Monte Carlo draws from a Bernoulli approximating variational distribution [64, 66]. Monte Carlo Dropout allows practitioners to compute the predictive mean and variance of multiple stochastic forward passes, yielding a highly reliable estimate of epistemic uncertainty without altering the standard deterministic training pipeline or artificially inflating the parameter count [65, 67].

Simultaneously, Deep Ensembles were established as a conceptually simple yet exceptionally formidable baseline for predictive uncertainty [23, 27]. The methodology involves training multiple identically architected neural networks independently, relying solely on the stochasticity of random weight initialization and randomized data shuffling to induce functional diversity across the ensemble [27]. Predictions are aggregated by calculating the mean of the ensemble components, and epistemic uncertainty is natively derived from the variance across the diverse predictive outputs.

Deep Ensembles routinely outperform strict Bayesian methods like Mean-Field Variational Inference in empirical benchmarks [23, 68]. From a strict Bayesian perspective, Deep Ensembles can be interpreted through the lens of particle-optimization variational inference, where each ensemble member represents a Dirac delta function approximating a distinct mode within a highly complex, multimodal posterior landscape [69, 70]. Furthermore, recent PAC-Bayesian frameworks theoretically validate Deep Ensembles by demonstrating that weighting ensemble members to minimize a tandem loss function—which explicitly penalizes correlations among predictors—provides non-vacuous bounds on generalization by capitalizing on the "cancellation of errors" effect [71, 72]. To actively prevent ensemble members from collapsing into identical functional representations, Repulsive Deep Ensembles inject a kernelized repulsive force during training, coercing the individual weight trajectories to explore diverse posterior modes in a manner analogous to Wasserstein gradient flows [69, 70].

Evidential Deep Learning

Evidential Deep Learning diverges completely from the paradigm of modeling uncertainty within the network's weight space. Instead, it shifts the probabilistic parameterization directly to the output space, grounding its architecture in the Dempster-Shafer Theory of Evidence and the principles of subjective logic [73, 74].

Instead of generating fixed point probabilities via a standard Softmax normalization layer, an Evidential Deep Learning network maps its extracted features to the concentration parameters of a Dirichlet distribution. The Dirichlet distribution functions as a meta-distribution—a probability distribution over all possible class probability assignments [74, 75]. For a multi-class classification problem, the network outputs a non-negative evidence vector, which is mathematically mapped to the Dirichlet parameters. When the network has accumulated substantial evidence for a specific class, the Dirichlet distribution narrows sharply around that classification. Conversely, when the total collected evidence is minimal, the network yields a flat, uniform Dirichlet distribution, explicitly outputting a state of total vacuity, or epistemic ignorance [73, 75].

Evidential Deep Learning guarantees ultra-fast, single-pass inference, entirely avoiding the heavy sampling costs associated with Markov Chain Monte Carlo, Monte Carlo Dropout, and Deep Ensembles. However, its reliance on conjugate priors and deterministic mapping functions introduces significant theoretical limitations. A growing body of rigorous literature critiques Evidential Deep Learning, highlighting that in asymptotic limits with infinite data, the model's distributional uncertainty often fails to vanish, thereby exposing deep structural flaws in its epistemic guarantees [76, 77]. To counteract these fundamental limitations, modern architectural adaptations apply Normalizing Flows or distill true Bayesian Neural Network responses directly into the Dirichlet priors, thereby anchoring the Evidential framework in verifiable Bayesian logic [76, 78].

Deterministic and Single-Pass Uncertainty Quantification

Expanding upon the critical requirement for single-forward-pass uncertainty without inheriting the theoretical restrictions of Evidential Deep Learning, several deterministic methodologies operate directly on the latent feature spaces of deep architectures.

Methods such as Spectral-normalized Neural Gaussian Processes (SNGP) substitute the final linear classification layer of a deep network with a distance-aware Gaussian Process, optimized via Random Fourier Features [79, 80]. Crucially, SNGP imposes a strict bi-Lipschitz constraint across the network's hidden layers utilizing spectral normalization. This architectural modification ensures mathematical distance awareness: if two inputs are vastly separated in the raw data space, they are guaranteed to remain far apart in the mapped hidden feature space [79]. Consequently, out-of-distribution test data residing far from the training manifold will securely yield high predictive variance in the terminal Gaussian Process [79, 80].

Alternative architectures, such as the Deep Vecchia Ensemble and Probabilistic Skip Connections, analyze Neural Collapse metrics to explicitly identify optimally smooth and sensitive intermediate network layers [81, 82]. By retrofitting deterministic uncertainty trackers directly into these identified layers within pre-trained models, these methods achieve state-of-the-art out-of-distribution detection without necessitating any modifications to the original, highly optimized training schemas [81, 83].

Kernel Methods and Deep Gaussian Processes

Bridging the theoretical gap between highly parametric neural networks and non-parametric statistical methods, Deep Gaussian Processes construct a hierarchical stack of Gaussian Process mappings [84, 85]. Rather than passing data through sequential deterministic weight matrices, Deep Gaussian Processes treat the output of one multivariate Gaussian Process as the direct input to the subsequent Gaussian Process in the hierarchy [84, 86]. While exact inference in this deeply nested structure is analytically intractable, Deep Gaussian Processes can be solved efficiently through variational approximations employing sparse inducing points, providing superior uncertainty quantification particularly in low-data regimes where standard neural networks overfit catastrophically [85, 87].

Additionally, the Neural Tangent Kernel theory establishes a profound mathematical equivalence between artificial neural networks and kernel methods. The theory dictates that, under the strict conditions of infinitesimal learning rates and an infinite-width limit, a neural

network trained via standard gradient descent behaves identically to kernel ridge regression leveraging a specific, deterministic kernel function [88, 89]. This infinite-width limit effectively forces the network to exhibit Gaussian Process behavior throughout its training evolution, providing deep theoretical linkages between initialization schemes, parameter gradient flow, and rigorous generalization bounds [88, 90].

Parameter-Efficient Bayesian Deep Learning for Large Language Models

As artificial intelligence advances inexorably into the era of billion-parameter Large Language Models and massive multi-modal foundation models, the deployment of strict Bayesian techniques has encountered severe computational walls. Performing exact Stochastic Gradient Markov Chain Monte Carlo or generating full-network Laplace Approximations over architectures containing seventy billion parameters is computationally and financially intractable [91, 92]. However, the notorious tendency for fine-tuned Large Language Models to "hallucinate" overconfidently on complex reasoning tasks makes the integration of rigorous Uncertainty Quantification an absolute necessity [93, 94].

Bayesian Low-Rank Adaptation

Parameter-Efficient Fine-Tuning methodologies, most prominently Low-Rank Adaptation (LoRA), address the parameter scale crisis by freezing the massive pre-trained base weights of the model and exclusively injecting trainable, low-rank matrices into specific network layers [11, 91]. Recent literature has elegantly mapped Bayesian inference algorithms directly onto this low-rank subspace, drastically compressing the dimensionality of the uncertainty tracking problem and making Bayesian Large Language Models a reality [92, 95].

Applying post-hoc Laplace approximations strictly over the Low-Rank Adaptation parameters entirely bypasses the need to compute intractable Hessians over the vast foundation model. By employing Kronecker-factored approximations exclusively on the low-rank parameter blocks, Laplace-LoRA significantly mitigates miscalibration and hallucinated overconfidence in question-answering tasks, incurring virtually no added latency during inference [92, 96].

Expanding beyond static Hessian curvature estimations, the Bayesian-LoRA framework establishes a profound structural isomorphism between deterministic Low-Rank Adaptation updates and Kronecker-factored Sparse Gaussian Process posteriors [11, 97]. Using variational inference augmented by normalizing flows within this restricted inducing space, Bayesian-LoRA optimizes the model's uncertainty profile end-to-end alongside the language model. This approach requires fewer than half a million additional parameters while achieving radical reductions in Expected Calibration Error across dense models as large as thirty billion parameters [11, 97, 98].

Further continuous integration is achieved via Bayesian LoRA by Backprop (BLoB). This algorithm continuously tracks both the mean and the covariance of the low-rank matrices using variational bounds throughout the entire instruction fine-tuning phase, preventing the dangerous decoupling of point estimation and uncertainty estimation that plagues post-hoc methods [93]. Maximizing parameter efficiency even further, ScalaBL (Stochastic Variational

Subspace Inference) performs Bayesian inference in an extreme r -dimensional subspace, repurposing the LoRA parameters as direct projection matrices. This breakthrough requires thousands of times fewer variational parameters than BLoB while scaling seamlessly to massive pre-trained architectures [95]. Adaptive methodologies like Bayesian Adaptive Rank Allocation extend this efficiency by dynamically allocating rank capacities per layer and context, imposing structured sparsity to enhance model calibration on complex, few-shot reasoning benchmarks [99].

Bayesian LLM Method	Core Mechanism	Parameter Efficiency	Key Advantage
Laplace-LoRA	Post-hoc Laplace on LoRA	Extremely High	No changes to training pipeline
Bayesian-LoRA (SGP)	Sparse Gaussian Process isomorphism	High	End-to-end uncertainty training
BLoB	Variational Inference on LoRA matrices	Moderate	Jointly updates mean and covariance
ScalaBL	Subspace inference via projection	Very High	Scales to 30B+ parameters easily

Bayesian Active Learning and Alignment

In the context of aligning Large Language Models with human intent via Reinforcement Learning from Human Feedback or Direct Preference Optimization, the acquisition of high-quality human preference data represents the primary operational bottleneck. Bayesian Active Learning specifically utilizes the epistemic uncertainty extracted from Bayesian-LoRA architectures to orchestrate highly intelligent, data-efficient acquisition policies [100, 101]. Systems such as the Bayesian Active Learner for Preference Modeling construct acquisition policies that deliberately query humans only on prompts exhibiting maximal epistemic uncertainty and maximal feature-space entropy. This ensures human raters are only forced to evaluate highly diverse and informative edge cases, rather than redundant, easily predictable data points [101, 102]. Empirical evaluations demonstrate that utilizing Bayesian active sampling reduces the required volume of preference labels by up to sixty-eight percent while strictly improving overall model alignment and safety [101]. Furthermore, treating reward modeling as a sequential Bayesian Experimental Design allows models to audit themselves dynamically, iteratively questioning human users to resolve ambiguous or unidentifiable reward landscapes

in a mathematically optimal sequence [100, 103].

Challenges and Prospects

While the theoretical mechanics of Bayesian Deep Learning are well established and increasingly scalable, translating these frameworks to unconstrained, real-world deployment faces several persistent anomalies and ongoing research hurdles.

The Cold Posterior Effect

The Cold Posterior Effect represents one of the most heavily scrutinized paradoxes in modern Bayesian Deep Learning. Theoretically, the exact Bayesian posterior corresponding to a standard temperature should unconditionally provide optimal predictive distributions [104, 105]. However, practitioners consistently observe that artificially "cooling" or sharpening the posterior by raising the likelihood to a fractional power significantly and universally improves empirical test accuracy [104, 106].

Cooling the posterior mathematically mimics overcounting the training data or drastically shrinking the prior's influence. Three dominant hypotheses currently attempt to explain this jarring divergence from Bayesian orthodoxy [104, 107]:

1. **Prior Misspecification:** Standard Bayesian Deep Learning routinely utilizes isotropic Gaussian priors over network weights, explicitly assuming all parameters act independently. This structural assumption does not align with the hierarchical, highly correlated geometry of deep networks, leading to poorly constrained hypothesis spaces that massively overestimate aleatoric uncertainty on otherwise clean datasets [105, 106].
2. **Likelihood Corruption via Data Augmentation:** Standard supervised learning relies heavily on manipulating data via synthetic augmentation, such as image cropping or flipping. When these synthetic augmentations are mathematically treated as independent and identically distributed observations in the strict likelihood calculation, the fundamental Bayesian assumption is broken, necessitating an ad-hoc temperature shift to realign the corrupted posterior [104, 108].
3. **Model Misspecification and Curation:** Neural networks inherently over-fit or under-fit specific components of the feature space due to architectural constraints. Tempering the posterior effectively provides a coarse, decision-theoretic generalized Bayes mechanism to artificially rebalance the misspecified loss surface [104, 107].

Solving the Cold Posterior Effect demands the construction of highly informative, structurally aware priors [104, 109]. Integrating domain-specific knowledge as informative priors, and utilizing knowledge-augmented adaptive likelihoods, demonstrates promising early success in preventing cold-posterior collapse by anchoring the hypothesis space to established logical constraints rather than relying on uninformative Gaussian assumptions [109, 110].

Scalability and End-to-End Workflows

The historical critique of Bayesian Deep Learning—that it is fundamentally unscalable and excessively slow—is rapidly dissolving. The synthesis of robust algorithmic enablers, such as Kronecker-factored curvature and matrix-valued kernels, with advanced hardware ecosystems and highly optimized compilers has effectively closed the computational gap [45, 54, 111].

Modern libraries permit practitioners to deploy intricate Markov Chain Monte Carlo and Laplace routines with near-identical wall-clock training times compared to standard deterministic training [45, 54].

Moving forward, the Bayesian Deep Learning community is heavily prioritizing "sample distillation"—compressing massive, memory-intensive sets of posterior samples into single, highly efficient forward-pass student models. Through techniques like Conformal Laplace-Aware Predictive Scaling or direct knowledge distillation over Bayesian outputs, models can completely remove test-time computational overhead while retaining mathematically valid, conformal uncertainty bounds [112, 113, 114].

Conclusion

As deep neural networks continue to migrate from isolated academic benchmarks into real-world autonomous systems, biomedical inference platforms, and ubiquitous natural language processing agents, the absolute demand for robust, transparent confidence estimation is paramount. Bayesian Deep Learning addresses the structural overconfidence inherent in standard point-estimated deep networks by enforcing probabilistic integration over all plausible models, allowing systems to explicitly quantify their own ignorance.

This survey demonstrates that while exact Bayesian inference in deep networks remains an intractable ideal, the ecosystem of approximate methodologies—ranging from Variational Inference and Markov Chain Monte Carlo to Deep Ensembles and Deterministic bounds—has matured considerably. Crucially, the recent fusion of Parameter-Efficient Fine-Tuning with Bayesian approximations has decisively breached the scalability barrier, enabling high-fidelity epistemic uncertainty quantification inside billion-parameter language models without requiring supercomputing clusters. By systematically addressing remaining anomalies, such as the Cold Posterior Effect, through the adoption of structured informative priors and generalized decision-theoretic loss bounds, Bayesian principles are poised to become the definitive standard for engineering reliable, safe, and trustworthy artificial intelligence.

References

- [1] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *International Conference on Machine Learning (ICML)*. arXiv:1706.04599.
- [2] Ovadia, Y., et al. (2019). Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [3] He, W., Jiang, Z., Xiao, T., Xu, Z., & Li, Y. (2023). A Survey on Uncertainty Quantification Methods for Deep Learning. *ACM Computing Surveys*. arXiv:2302.13425.
- [4] Abdar, M., et al. (2021). A Review of Uncertainty Quantification in Deep Learning: Techniques, Applications and Challenges. *Information Fusion*.
- [5] van Leeuwen, P. V. (2024). Uncertainty Quantification for Deep Learning. *Environmental Data Science*. arXiv:2405.20550.
- [6] Gawlikowski, J., et al. (2021). A Survey of Uncertainty in Deep Neural Networks. *Artificial Intelligence Review*.
- [7] Blundell, C., Cornebise, J., Kavukcuoglu, K., & Wierstra, D. (2015). Weight Uncertainty in Neural Networks. *International Conference on Machine Learning (ICML)*. arXiv:1505.05424.

- [8] Wang, H., & Ji, Q. (2020). Bayesian Deep Learning: A Survey. *ACM Computing Surveys*. arXiv:1604.01662.
- [9] Papamarkou, T., et al. (2024). Position: The Time for Sampling Is Now! Charting a New Course for Bayesian Deep Learning. *International Conference on Machine Learning (ICML)*. arXiv:2605.21765.
- [10] Jospin, L. V., et al. (2022). Hands-on Bayesian Neural Networks - A Tutorial for Deep Learning Users. *IEEE Computational Intelligence Magazine*.
- [11] Yang, A. X., Robeyns, M., Wang, X., & Aitchison, L. (2024). Bayesian Low-rank Adaptation for Large Language Models. *International Conference on Learning Representations (ICLR)*. arXiv:2308.13111.
- [12] Lin, M., et al. (2026). Bayesian-LoRA: Probabilistic Low-Rank Adaptation of Large Language Models. *Conference on Neural Information Processing Systems (NeurIPS)*. arXiv:2601.21003.
- [13] Wang, X., et al. (2024). Bayesian Low-Rank Adaptation by Backpropagation (BLoB). *Conference on Neural Information Processing Systems (NeurIPS)*.
- [14] Kendall, A., & Gal, Y. (2017). What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1703.04977.
- [15] Hüllermeier, E., & Waegeman, W. (2021). Aleatoric and Epistemic Uncertainty in Machine Learning: An Introduction to Concepts and Methods. *Machine Learning*.
- [16] Der Kiureghian, A., & Ditlevsen, O. (2009). Aleatory or epistemic? Does it matter? *Structural Safety*.
- [17] Gruber, S., et al. (2023). Sources of Uncertainty in Machine Learning: A Statisticians' View. *Wiley Interdisciplinary Reviews: Computational Statistics*.
- [18] Malinin, A., & Gales, M. (2018). Predictive Uncertainty Estimation via Prior Networks. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [19] Ruff, L., et al. (2021). A Unifying Review of Deep and Shallow Anomaly Detection. *Proceedings of the IEEE*.
- [20] MacKay, D. J. C. (1992). A Practical Bayesian Framework for Backpropagation Networks. *Neural Computation*.
- [21] Neal, R. M. (1996). Bayesian Learning for Neural Networks. *Lecture Notes in Statistics*. Springer.
- [22] Wilson, A. G., & Izmailov, P. (2020). Bayesian Deep Learning and a Probabilistic Perspective of Generalization. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [23] Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1612.01474.
- [24] Naeini, M. P., Cooper, G. F., & Hauskrecht, M. (2015). Obtaining Well Calibrated Probabilities Using Bayesian Binning. *AAAI Conference on Artificial Intelligence*.
- [25] Nixon, J., Dusenberry, M. W., Zhang, L., Jerfel, G., & Tran, D. (2019). Measuring Calibration in Deep Learning. *CVPR Workshops*.
- [26] Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*.
- [27] Gneiting, T., & Raftery, A. E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation.

Journal of the American Statistical Association.

- [28] Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational Inference: A Review for Statisticians. *Journal of the American Statistical Association.*
- [29] Hinton, G. E., & van Camp, D. (1993). Keeping the neural networks simple by minimizing the description length of the weights. *Proceedings of the sixth annual conference on Computational learning theory (COLT).*
- [30] Graves, A. (2011). Practical Variational Inference for Neural Networks. *Advances in Neural Information Processing Systems (NeurIPS).*
- [31] Bai, J., Song, Q., & Cheng, G. (2020). Efficient Variational Inference for Sparse Deep Learning with Theoretical Guarantee. *Advances in Neural Information Processing Systems (NeurIPS).*
- [32] Monemi, M., Amini, M., Taheri, S. M., & Arashi, M. (2025). On weight and variance uncertainty in neural networks for regression tasks. *Machine Learning.* arXiv:2501.04272.
- [33] Kristiadi, A., Hein, M., & Hennig, P. (2020). Being Bayesian, Even Just a Bit, Fixes Overconfidence in ReLU Networks. *International Conference on Machine Learning (ICML).*
- [34] Harrison, J., et al. (2024). Variational Bayesian Last Layers. *International Conference on Learning Representations (ICLR).*
- [35] Watson, J., et al. (2021). Latent Derivative Bayesian Last Layer Networks. *International Conference on Artificial Intelligence and Statistics (AISTATS).*
- [36] Robert, C. P., & Casella, G. (2004). Monte Carlo Statistical Methods. *Springer.*
- [37] Welling, M., & Teh, Y. W. (2011). Bayesian Learning via Stochastic Gradient Langevin Dynamics. *International Conference on Machine Learning (ICML).*
- [38] Nemeth, C., & Fearnhead, P. (2021). Stochastic Gradient Markov Chain Monte Carlo. *Journal of the American Statistical Association.*
- [39] Teh, Y. W., Thiery, A. H., & Vollmer, S. J. (2016). Consistency and fluctuations for stochastic gradient Langevin dynamics. *Journal of Machine Learning Research (JMLR).*
- [40] Zhang, R., et al. (2022). Low-Precision Stochastic Gradient Langevin Dynamics. *International Conference on Machine Learning (ICML).*
- [41] Chen, T., Fox, E., & Guestrin, C. (2014). Stochastic Gradient Hamiltonian Monte Carlo. *International Conference on Machine Learning (ICML).*
- [42] Ding, N., et al. (2014). Bayesian Sampling Using Stochastic Gradient Thermostats. *Advances in Neural Information Processing Systems (NeurIPS).*
- [43] Zhang, R., Li, C., Zhang, J., Chen, C., & Wilson, A. G. (2020). Cyclical Stochastic Gradient MCMC for Bayesian Deep Learning. *International Conference on Learning Representations (ICLR).*
- [44] Chen, C., et al. (2016). Bridging the Gap between Stochastic Gradient MCMC and Stochastic Optimization. *International Conference on Artificial Intelligence and Statistics (AISTATS).*
- [45] Cabezas, A., et al. (2024). blackjax: A sampling library for JAX. *Journal of Open Source Software.*
- [46] Duffield, S., et al. (2025). JaxSGMC: A JAX library for Stochastic Gradient MCMC. *SoftwareX.* arXiv:2505.11190.
- [47] Liu, Q., & Wang, D. (2016). Stein Variational Gradient Descent: A General Purpose Bayesian

- Inference Algorithm. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1608.04471.
- [48] Gorham, J., & Mackey, L. (2015). Measuring Sample Quality with Stein's Method. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [49] Liu, Q. (2017). Stein Variational Gradient Descent as Gradient Flow. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [50] Wang, D., Tang, Z., Bajaj, C., & Liu, Q. (2019). Stein Variational Gradient Descent With Matrix-Valued Kernels. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1910.12794.
- [51] D'Angelo, F., & Fortuin, V. (2021). Annealed Stein Variational Gradient Descent. *Symposium on Advances in Approximate Bayesian Inference (AABI)*.
- [52] Ba, J., et al. (2023). Variance Collapse in Stein Variational Gradient Descent. *International Conference on Machine Learning (ICML)*. arXiv:2305.10636.
- [53] Grathwohl, W., et al. (2020). Neural Variational Gradient Descent. *International Conference on Learning Representations (ICLR)*.
- [54] Daxberger, E., Kristiadi, A., Immer, A., Eschenhagen, R., Bauer, M., & Hennig, P. (2021). Laplace Redux – Effortless Bayesian Deep Learning. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [55] Eschenhagen, R., et al. (2025). laplax: A modular library for Laplace approximations in JAX. *International Conference on Learning Representations (ICLR)*.
- [56] Ritter, H., Botev, A., & Barber, D. (2018). A Scalable Laplace Approximation for Neural Networks. *International Conference on Learning Representations (ICLR)*.
- [57] Immer, A., Korzepa, M., & Bauer, M. (2021). Improving predictions of Bayesian neural nets via local linearization. *International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- [58] Martens, J., & Grosse, R. (2015). Optimizing Neural Networks with Kronecker-factored Approximate Curvature. *International Conference on Machine Learning (ICML)*.
- [59] Botev, A., Ritter, H., & Barber, D. (2017). Practical Gauss-Newton Optimisation for Deep Learning. *International Conference on Machine Learning (ICML)*.
- [60] Antorán, J., et al. (2022). Adapting the Linearised Laplace Model Evidence for Modern Deep Learning. *International Conference on Machine Learning (ICML)*.
- [61] Kristiadi, A., et al. (2021). Learnable Uncertainty under Laplace Approximations. *Uncertainty in Artificial Intelligence (UAI)*.
- [62] Daxberger, E., et al. (2026). Subnetwork Laplace: Streamlined Predictive Uncertainty via Local Linearization. *International Conference on Learning Representations (ICLR)*. arXiv:2603.29466.
- [63] Wolinski, M., & Arbel, M. (2022). Local Gaussian approximations for fast predictive distributions in Bayesian Neural Networks. *Symposium on Advances in Approximate Bayesian Inference (AABI)*.
- [64] Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *International Conference on Machine Learning (ICML)*. arXiv:1506.02142.
- [65] Gal, Y., & Ghahramani, Z. (2016). A Theoretically Grounded Application of Dropout in Recurrent Neural Networks. *Advances in Neural Information Processing Systems (NeurIPS)*.

- [66] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research (JMLR)*.
- [67] Kingma, D. P., Salimans, T., & Welling, M. (2015). Variational Dropout and the Local Reparameterization Trick. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [68] Ovadia, Y., et al. (2019). Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [69] D'Angelo, F., & Fortuin, V. (2021). Repulsive Deep Ensembles are Bayesian. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [70] Fort, S., Hu, H., & Lakshminarayanan, B. (2019). Deep Ensembles: A Loss Landscape Perspective. *arXiv preprint arXiv:1912.02757*.
- [71] Masegosa, A. R., et al. (2020). Second Order PAC-Bayesian Bounds for the Weighted Majority Vote. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [72] Ortega, L., et al. (2022). PAC-Bayesian Deep Ensembles. *International Conference on Machine Learning (ICML)*. arXiv:2406.05469.
- [73] Sensoy, M., Kaplan, L., & Kandemir, M. (2018). Evidential Deep Learning to Quantify Classification Uncertainty. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [74] Jøsang, A. (2016). Subjective Logic: A Formalism for Reasoning Under Uncertainty. *Springer*.
- [75] Amini, A., Schwarting, W., Soleimany, A., & Rus, D. (2020). Deep Evidential Regression. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [76] Bengs, V., et al. (2022). Pitfalls of Epistemic Uncertainty Quantification through Loss Minimisation. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [77] Meinert, D., & Lavin, A. (2023). Uncovering the Limits of Evidential Deep Learning. *International Conference on Machine Learning (ICML)*.
- [78] Wang, H., & Ji, Q. (2024). Beyond Dirichlet-based Models: When Bayesian Neural Networks Meet Evidential Deep Learning. *International Conference on Machine Learning (ICML)*.
- [79] Liu, J. Z., Lin, Z., Padhy, S., Tran, D., Bedrax-Weiss, T., & Balaji, L. (2020). Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [80] van Amersfoort, J., Smith, L., Teh, Y. W., & Gal, Y. (2020). Uncertainty Estimation Using a Single Deep Deterministic Neural Network. *International Conference on Machine Learning (ICML)*.
- [81] Mukhoti, J., Kirsch, A., van Amersfoort, J., Torr, P. H., & Gal, Y. (2023). Deep Deterministic Uncertainty: A Simple Baseline. *International Conference on Learning Representations (ICLR)*.
- [82] Pappayan, V., Trinh, M. C., & Sulam, D. (2020). Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences (PNAS)*.
- [83] Postels, J., et al. (2025). Probabilistic Skip Connections for Single-Pass Deterministic Uncertainty. *International Conference on Learning Representations (ICLR)*. arXiv:2501.04816.
- [84] Damianou, A., & Lawrence, N. D. (2013). Deep Gaussian Processes. *International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- [85] Salimbeni, V., & Deisenroth, M. (2017). Doubly Stochastic Variational Inference for Deep

- Gaussian Processes. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [86] Cutajar, K., et al. (2017). Random Feature Expansions for Deep Gaussian Processes. *International Conference on Machine Learning (ICML)*.
- [87] Bui, T. D., et al. (2016). Deep Gaussian Processes for Regression using Approximate Expectation Propagation. *International Conference on Machine Learning (ICML)*.
- [88] Jacot, A., Gabriel, F., & Hongler, C. (2018). Neural Tangent Kernel: Convergence and Generalization in Neural Networks. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1806.07572.
- [89] Lee, J., et al. (2019). Wide Neural Networks of Any Depth Evolve as Linear Models Under Gradient Descent. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [90] Arora, S., et al. (2019). On Exact Computation with an Infinitely Wide Neural Net. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [91] Hu, E. J., et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. *International Conference on Learning Representations (ICLR)*.
- [92] Daxberger, E., et al. (2024). Laplace-LoRA: Bayesian Low-Rank Adaptation for Large Language Models. *International Conference on Learning Representations (ICLR)*.
- [93] OpenAI. (2023). GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*.
- [94] Kadavath, S., et al. (2022). Language Models (Mostly) Know What They Know. *arXiv preprint arXiv:2207.05221*.
- [95] Zhang, Y., et al. (2025). Scalable Bayesian Low-Rank Adaptation via Stochastic Variational Subspace Inference (ScalaBL). *International Conference on Learning Representations (ICLR)*. arXiv:2506.21408.
- [96] Antorán, J., et al. (2022). Adapting the Linearised Laplace Model Evidence for Modern Deep Learning. *International Conference on Machine Learning (ICML)*.
- [97] Titsias, M. K. (2009). Variational Learning of Inducing Variables in Sparse Gaussian Processes. *International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- [98] Burt, D. R., Rasmussen, C. E., & van der Wilk, M. (2020). Convergence of Sparse Variational Inference in Gaussian Processes Regression. *Journal of Machine Learning Research (JMLR)*.
- [99] Wang, X., et al. (2026). Bayesian Adaptive Rank Allocation (BaRA) for Parameter-Efficient Fine-Tuning. *International Conference on Learning Representations (ICLR)*. arXiv:2606.29184.
- [100] Ouyang, L., et al. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [101] Xu, Z., et al. (2024). Deep Bayesian Active Learning for Preference Modeling in Large Language Models. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2406.10023.
- [102] Rafailov, R., et al. (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [103] Bertolazzi, M., et al. (2025). Sequential Bayesian Experimental Design with Large Language Models (BED-LLM). *International Conference on Machine Learning (ICML)*. arXiv:2508.21184.
- [104] Wenzel, F., et al. (2020). How Good is the Bayes Posterior in Deep Neural Networks? *International Conference on Machine Learning (ICML)*.
- [105] Fortuin, V., Garriga-Alonso, A., Ober, S. W., Wenzel, F., Rätsch, G., Turner, R. E., ... & Aitchison, L. (2022). Bayesian Neural Network Priors Revisited. *International Conference on*

Learning Representations (ICLR).

[106] Aitchison, L. (2020). A Statistical Theory of Cold Posteriors in Deep Neural Networks. *International Conference on Learning Representations (ICLR).*

[107] Izmailov, P., et al. (2021). What Are Bayesian Neural Network Posteriors Really Like? *International Conference on Machine Learning (ICML).*

[108] Noci, L., et al. (2021). The Cold Posterior Effect in Bayesian Neural Networks: A Data Augmentation Perspective. *Advances in Neural Information Processing Systems (NeurIPS).*

[109] Ma, Y., et al. (2026). Towards Knowledge-augmented Bayesian Deep Learning For Computer Vision. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[110] Zhang, Y. (2024). On Cold Posteriors of Probabilistic Neural Networks. *Ph.D. Thesis, University of Copenhagen.*

[111] Bradbury, J., et al. (2018). JAX: composable machine learning and AI research. *GitHub repository.*

[112] Barber, D., et al. (2025). Conformal Laplace-Aware Predictive Scaling (CLAPS). *International Conference on Machine Learning (ICML).* arXiv:2512.01384.

[113] Angelopoulos, A. N., & Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *arXiv preprint arXiv:2107.07511.*

[114] Romano, Y., Patterson, E., & Candès, E. (2019). Conformalized Quantile Regression. *Advances in Neural Information Processing Systems (NeurIPS).*